EHIF (External Host Interface)
Design Specification

Ver 0.0

DM270

Digital Still Camera

Department: WW IAG

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HISTORY

Version	Date	Author	Notes
Ver 0.0			

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1. ABSTRACT

This module allows a host device to access AIM, DSP Internal memory and SDRAM connected to DM270.

2. REFERENCES

[DM270] MTC Design Specification
[DM270] AIM Design Specification Rev.0.1
[DM270] HPIB Design Specification Rev.0.1

3. GLOSSARY

EHCIF	Host Interface
BUSC	Bus Controller
ARM	ARM71TDMIE
SDRC	SDRAM Controller
USB	Universal Serial BUS
HPIB	Host Peripheral Interface Bridge

4. FUNCTION

4.1 Block Diagram

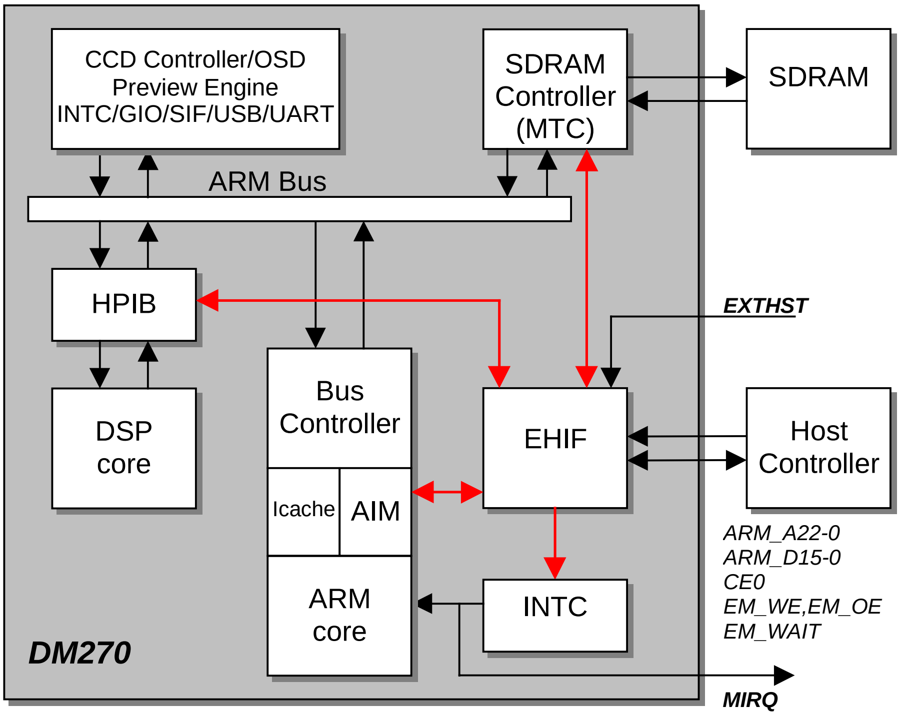

Figure 1 : Block Diagram describing Host Interface mode

By setting thepin EXTHST to high level, a host device can access some of the memory areas of DM270. When in external CPU mode, ARM_A22-ARM_A0, EM_CE0, EM_WE and EM_OE become input pins. The data bus is fixed at 16-bits and uses ARM_D15-ARM_D0. Access from a host device can be performed dynamically by enabling EM_WAIT output for the host device to monitor the status of the DM270. DM270 can interrupt the host device through GIO25, after GIO25 has been configured to function as the MIRQ (interrupt) signal.

4.2 Memory Space

The memory areas that can be accessed from an external host controller are ARM internal memory (32 Kbytes), SDRAM (16,128Kbytes), and DSP internal memory (128 Kbytes). However, the memory map addresses differ from the usual ARM memory map. Table 1 shows the memory map of DM270 as seen from the external CPU.

Table 1 : Memory Space

Memory Area	Start address	End address	size(byte)
EHIF registers	0x00:0000	0x00:007F	128
ARM Internal Memory	0x00:8000	0x00:FFFF	32 K
DSP Internal Memory	0x02:0000	0x03:FFFF	128 K
SDRAM	0x04:0000	0xFF:FFFF	16,128 K

4.3 Spec

- The data bus is fixed at 16-bits and uses ARM_D15-ARM_D0. Both read and write are half-word (16 bit) access.
- Access from a host device can be performed dynamically by enabling EM_WAIT output for the host device to monitor the status of the DM270 ('0':Busy, '1':ready). WAITEN register of the External CPU Interface must be set in advance to enable the EM_WAIT output.
- The access time seen from the host device varies depending on ARM, DSP and SDRAM clock frequency settings. (Refer to 4.4 Access Time)
- 32byte Burst/Continuous Burst access to SDRAM is supported via sixteen 16bit data registers.
- Host device can issue an interrupt signal to the ARM by setting the register INT.
- If a read is performed continuously after read, EM_OE or EM_CE0 must once return to high level. RTRG register must be set in case EM_OE becomes high level and EM_CE0 stays low.

4.4 Access Time

The External Host I/F module operates by the ARM clock of the DM270. The access time seen from the host device varies depending on ARM, DSP and SDRAM clock frequency settings. The number of cycles of the ARM clock is shown below.

EHIF register/SDRAM burst access/AIM write	: $2 + \alpha$ cycles	(α :Input and/or Output delay)
AIM read	: $3 + \alpha$ cycles	
SDRAM single access/DSP memory	: $3 + n + \alpha$ cycles (n:Peripheral wait cycles)	

4.5 SDRAM access

Single/32byte Burst/Continuous Burst access to SDRAM is supported. Sixteen 16bit data registers are used for Burst/Continuous Burst access. When in Continuous Burst transfer mode, data of the specified length in 32 byte unit are transferred to and from SDRAM. There are 2 modes for Continuous Burst access.

- **mode1** : Address decode enabled. Register, AIM and HPIB accesses are allowed during Continuous Burst access. There are some registers to monitor the status of EHIF during Continuous Burst access which are:
  1. Burst transfer wait flag : To check the Burst transfer status every 16 accesses. This is for a host device which does not have a WAIT input.
  2. SDRAM access wait flag : To check the access status between EHIF and SDRC before you start access to AIM/HPIB during Continuous Burst transfer, or before you end the Continuous Burst transfer.
- **Mode2** : No address decode. No other access is allowed during DMA.

4.6 Timing

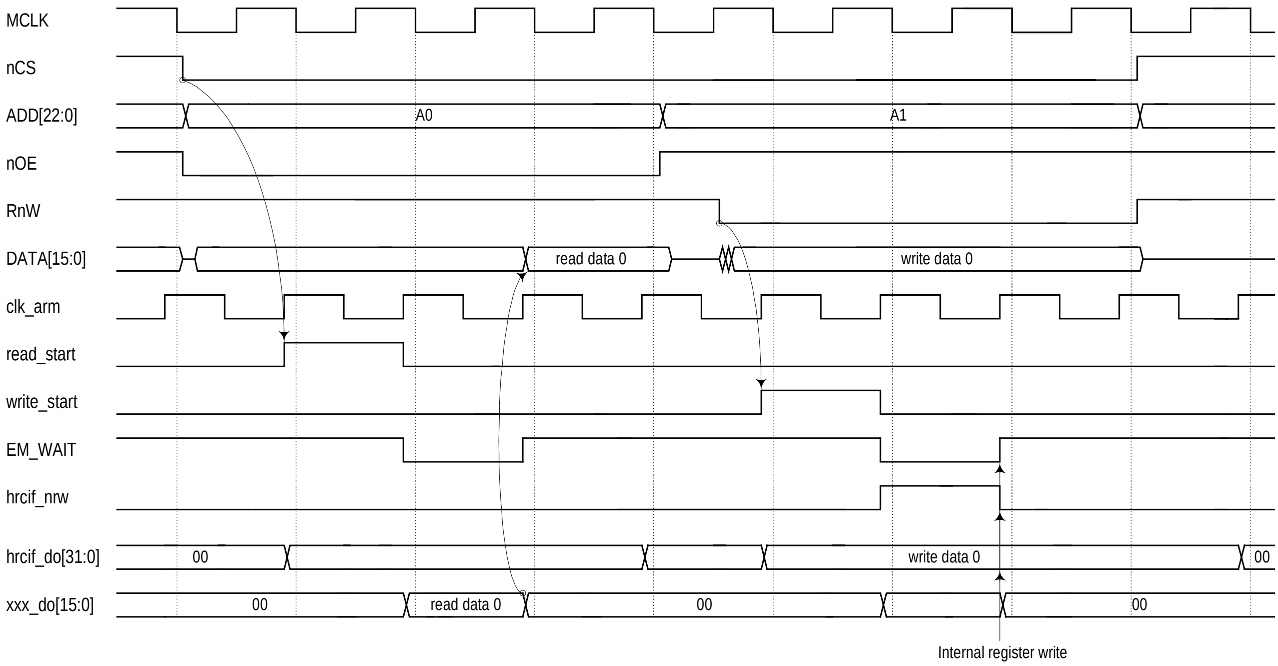

Figure 2 : External Host Interface for AIM

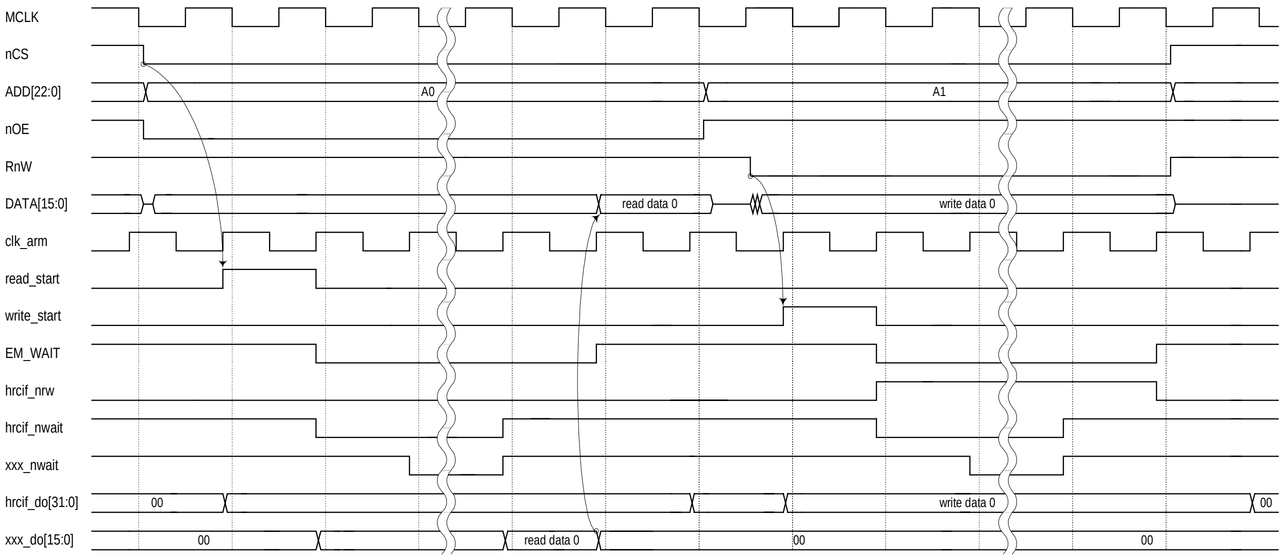

Figure 3 : External Host Interface for hand shake peripherals (HPIB)

5. REGISTERS

Table 3 : HRCIF Register List

Register Name	Offset address	Description
REVR	0x00	Device revision ID
EHIFCTL	0x01	EHIF control signals
INTCTL	0x02	Interrupt control
BRSTCTL1	0x03	SDRAM Burst transfer control 1
BRSTCTL2	0x04	SDRAM Burst transfer control 2
BRSTLEN	0x05	SDRAM Burst transfer length
SDRADR1	0x06	SDRAM address for Burst transfer
SDRADR2	0x07	SDRAM address for Burst transfer
SDRBUF0	0x08	Data registers for SDRAM Burst transfer
SDRBUF1	0x09	Data registers for SDRAM Burst transfer
SDRBUF2	0x0A	Data registers for SDRAM Burst transfer
SDRBUF3	0x0B	Data registers for SDRAM Burst transfer
SDRBUF4	0x0C	Data registers for SDRAM Burst transfer
SDRBUF5	0x0D	Data registers for SDRAM Burst transfer
SDRBUF6	0x0E	Data registers for SDRAM Burst transfer
SDRBUF7	0x0F	Data registers for SDRAM Burst transfer
SDRBUF8	0x10	Data registers for SDRAM Burst transfer
SDRBUF9	0x11	Data registers for SDRAM Burst transfer
SDRBUF10	0x12	Data registers for SDRAM Burst transfer
SDRBUF11	0x13	Data registers for SDRAM Burst transfer
SDRBUF12	0x14	Data registers for SDRAM Burst transfer
SDRBUF13	0x15	Data registers for SDRAM Burst transfer
SDRBUF14	0x16	Data registers for SDRAM Burst transfer
SDRBUF15	0x17	Data registers for SDRAM Burst transfer
EHIFSTS	0x18	EHIF status
TEST	0x02	Register for internal use

5.1 Detailed Register

REVR (0000:0000) Offset : 0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	REV7	REV6	REV5	REV4	REV3	REV2	REV1	REV0
R/W	-	-	-	-	-	-	-	-	R	R	R	R	R	R	R	R
Default	-	-	-	-	-	-	-	-	0	0	0	1	0	0	0	0

Bits	Register Name	Description
15:8	RSV	Reserved bit.
7:0	REV7-REV0	Device revision ID. eg.) ES1.0 => 0x10 Revision ID of DM270 can be read out from this register.

EHIFCTL (0000:0002) Offset : 0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	ENDIW1	ENDIW0	ENDIR1	ENDIR0	RSV	RSV	RSV	RTRG	RSV	RSV	RSV	WAITEN
R/W	-	-	-	-	R/W	R/W	R/W	R/W	-	-	-	R/W	-	-	-	R/W
Default	-	-	-	-	0	0	0	0	-	-	-	0	-	-	-	0

Bits	Register Name	Description
15:12	RSV	Reserved bit.
11:10	ENDIW	Endian select for Write access Bit0: byte swap enable Bit1: half-word swap enable(AIM, SDRAM)
9:8	ENDIR	Endian select for Read access Bit0: byte swap enable Bit1: half-word swap enable(AIM, SDRAM)
7:5	RSV	Reserved bit.
4	RTRG	Data Read trigger select bit. 0: nCS falling edge 1: nOE falling edge RTRG selects the trigger to start a Read access. A Read access is triggered by the CE0 falling edge in case RTRG='0', or by the EM_OE falling edge in case RTRG='1'.
3:1	RSV	Reserved bit.
0	WAITEN	EM_WAIT output enable. 0: EM_WAIT='Z' 1: EM_WAIT=OUTPUT(Busy/Ready status) Host device can monitor the status of DM270 through EM_WAIT. It allows the host device to dynamically access DM270.

INTCTL(0000:0004) Offset : 0x02

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	INT
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit.
0	INT	Issue an interrupt to ARM Writing '1' to this bit issues an interrupt to ARM. This bit will be cleared automatically.

BRSTCTL1 (0000:0006) Offset : 0x03

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	BFCLR	RSV	RSV	BRSTWR	BRSTRD
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	-	0	0

Bits	Register Name	Description
15:5	RSV	Reserved bit.
4	BFCLR	SDRAM data buffer clear. Writing '1' to this bit clears the content of SDRBUF[15:0]. This bit will be cleared automatically.
3:1	RSV	Reserved bit.
1	BRSTWR	SDRAM Burst Write enable. Writing '1' to this bit starts 32byte Burst Write transfer from SDRBUF[15:0] to SDRAM. This bit will be automatically cleared when the transfer is completed.
0	BRSTRD	SDRAM Burst Read enable. Writing '1' to this bit starts 32byte Burst Read transfer from SDRAM to SDRBUF[15:0]. This bit will be automatically cleared when the transfer is completed.

BRSTCTL2 (0000:0008) Offset : 0x04

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	CBRMD	RSV	RSV	CBRSWR	CBRSRD
R/W	-	-	-	-	-	-	-	-	-	-	-	R/W	-	-	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	0	-	-	0	0

Bits	Register Name	Description
15:5	RSV	Reserved bit.
4	CBRSMD	Specify a mode of SDRAM Continuous Burst transfer. 0: mode1 – Burst transfer will be performed by a writing or reading to/from a data register SDRBUF0. 1: mode2 – Address decoding will not be performed and every access from a host device will be treated as a burst transfer request once the data transfer starts, and until transfer of the spcified length ends.
3:1	RSV	Reserved bit.
1	CBRSWR	SDRAM Continuous Burst Write enable. Writing '1' to this bit starts Continuous Burst Write transfer of the spcified length set in BRSTLEN. Writing '0' to this bit ends data transfer, or otherwise, it will be automatically cleared when the transfer is completed.
0	CBRSRD	SDRAM Continuous Burst Read enable. Writing '1' to this bit starts Continuous Burst Read transfer of the spcified length set in BRSTLEN. Writing '0' to this bit ends data transfer, or otherwise, it will be automatically cleared when the transfer is completed.

BRSTLEN (0000:000A) Offset : 0x05

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	BRSTL15	BRSTL14	BRSTL13	BRSTL12	BRSTL11	BRSTL10	BRSTL9	BRSTL8	BRSTL7	BRSTL6	BRSTL5	BRSTL4	BRSTL3	BRSTL2	BRSTL1	BRSTL0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	BRSTL	Burst Length.

SDRADR1 (0000:000C) Offset : 0x06

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AINC	RSV	RSV	SDRA21	SDRA20	SDRA19	SDRA18	SDRA17	SDRA16
R/W	-	-	-	-	-	-	-	R/W	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	0	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15:9	RSV	Reserved bit.
8	AINC	SDRAM Address auto increment. Writing '1' to this bit enables auto address increment after the 1 shot burst transfer.
7:6	RSV	Reserved bit.
5:0	SDRA(21:16)	Upper 6 bits of SDRAM Address for burst transfer. Address is the offset from the SDRAM base and is specified in 32byte units.

SDRADR2 (0000:000E) Offset : 0x07

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDRA15	SDRA14	SDRA13	SDRA12	SDRA11	SDRA10	SDRA9	SDRA8	SDRA7	SDRA6	SDRA5	SDRA4	SDRA3	SDRA2	SDRA1	SDRA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDRA(15:0)	Lower 16 bits of SDRAM Address for burst transfer. Address is the offset from the SDRAM base and is specified in 32byte units.

SDRBUF0 (0000:0010) Offset : 0x08

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB0	Buffer0 for SDRAM burst transfer

SDRBUF1 (0000:0012) Offset : 0x09

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB115	SDB114	SDB113	SDB112	SDB111	SDB110	SDB19	SDB18	SDB17	SDB16	SDB15	SDB14	SDB13	SDB12	SDB11	SDB10
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB1	Buffer1 for SDRAM burst transfer

SDRBUF2 (0000:0014) Offset : 0x0A

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB215	SDB214	SDB213	SDB212	SDB211	SDB210	SDB29	SDB28	SDB27	SDB26	SDB25	SDB24	SDB23	SDB22	SDB21	SDB20
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB2	Buffer2 for SDRAM burst transfer

SDRBUF3 (0000:0016) Offset : 0x0B

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB315	SDB314	SDB313	SDB312	SDB311	SDB310	SDB39	SDB38	SDB37	SDB36	SDB35	SDB34	SDB33	SDB32	SDB31	SDB30
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB3	Buffer3 for SDRAM burst transfer

SDRBUF4 (0000:0018) Offset : 0x0C

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB4	Buffer4 for SDRAM burst transfer

SDRBUF5 (0000:001A) Offset : 0x0D

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description

Bits	Register Name	Description
15:0	SDB5	Buffer5 for SDRAM burst transfer

SDRBUF6 (0000:001C) Offset : 0x0E

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB6	Buffer6 for SDRAM burst transfer

SDRBUF7 (0000:001E) Offset : 0x0F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB7	Buffer7 for SDRAM burst transfer

SDRBUF8 (0000:0020) Offset : 0x10

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB8	Buffer8 for SDRAM burst transfer

SDRBUF9 (0000:0022) Offset : 0x11

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB9	Buffer9 for SDRAM burst transfer

SDRBUF10 (0000:0024) Offset : 0x12

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB10	Buffer10 for SDRAM burst transfer

SDRBUF11 (0000:0026) Offset : 0x13

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB12	Buffer11 for SDRAM burst transfer

SDRBUF12 (0000:0028) Offset : 0x14

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB12	Buffer12 for SDRAM burst transfer

SDRBUF13 (0000:002A) Offset : 0x15

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB13	Buffer13 for SDRAM burst transfer

SDRBUF14 (0000:002C) Offset : 0x16

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB14	Buffer14 for SDRAM burst transfer

SDRBUF15 (0000:002E) Offset : 0x17

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SDB015	SDB014	SDB013	SDB012	SDB011	SDB010	SDB09	SDB08	SDB07	SDB06	SDB05	SDB04	SDB03	SDB02	SDB01	SDB00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SDB15	Buffer15 for SDRAM burst transfer

EHIFSTS (0000:0030) Offset : 0x18

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	PLLLK	RSV	RSV	RSV	SDRBSY	RSV	RSV	RSV	BRSWT
R/W	-	-	-	-	-	-	-	R	-	-	-	R	-	-	-	R
Default	-	-	-	-	-	-	-	0	-	-	-	0	-	-	-	0

Bits	Register Name	Description
15:9	RSV	Reserved bit.
8	PLLLK	PLL lock status ‘1’ at this bit indicates PLL is locked.
7:5	RSV	Reserved bit.
4	SDRBSY	SDRAM Access busy flag. ‘1’ at this bit indicates that Data transfer between EHIF and SDRC is busy. HPIB and AIM can be accessed during SDRAM Continuous Burst transfer (mode0) after this bit becomes ‘0’.
3:1	RSV	Reserved bit.
0	BRSWT	Burst transfer access wait flag ‘1’ at this bit indicates that wait state is asserted for Burst transfer access, so that any access to the data register (SDRBUF) is prohibited.

TEST (0000:007E) Offset : 0x3F

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	AIMWT
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved bit.
0	AIMWT	AIM wait signal enable

6. AC TIMING

6.1 READ

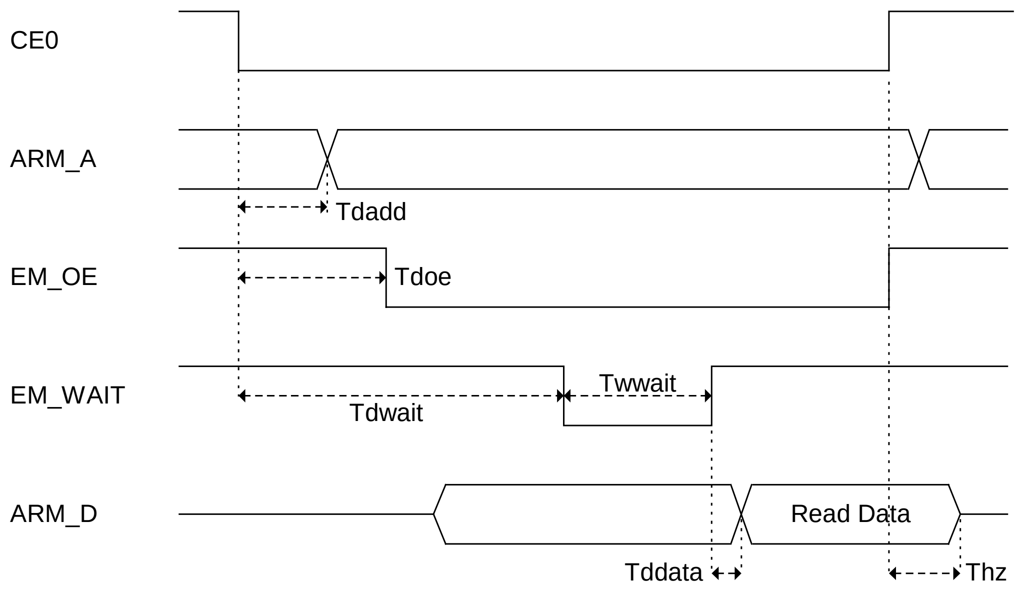

Figure 4. External Host Interface Read Timing

Name	Min (ns)	Max (ns)
Tdoe	--	Tcyc - 4
Tdadd	--	Tcyc - 8
Tdwait	--	Tcycx2 + 20
Twwait	Tcyc - 2	--
Tddata	--	3
Thz	0	12

Tcyc = DM270 System clock cycle period

6.2 WRITE

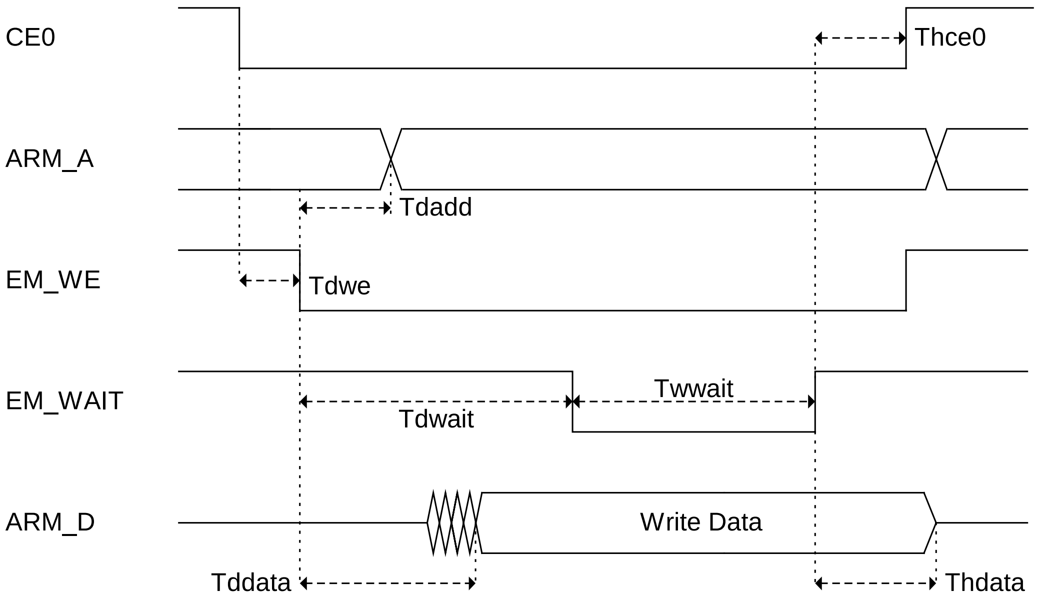

Figure 5. External Host Interface Write Timing

Name	Min (ns)	Max (ns)
Tdwe	0	--
Tdadd	--	Tcyc – 8
Tdwait	--	Tcycx2 + 20
Twwait	Tcyc - 2	--
Tddata	--	Tcyc - 4
Thdata	0	--
Thce0	0	--

Tcyc = DM270 System clock cycle period